VITyarthi | Open Source Software (CSE 0002)

The Open Source Audit

A Capstone Project for OSS NGMC Course

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INTRODUCTION

Linux and Free, Open Source Software (FOSS) are part of modern computing. Open source software offers a chance for programmers to access, modify, and share source code freely, which helps programmers to be innovative and collaborative. Some of the common software that can be classified as FOSS include Python, a high- level programming language that is famous for its clean and readable code. Web programming, data science, AI, and automation can be done using Python. It is open source, meaning that anyone can access the source code of Python and modify it as they wish. A very large and global community of support, mainly used for scripting and development of Linux systems.

This project aims to give information pertaining to the major concepts of Linux and FOSS (Free & Open Source Software) in open source software on Python and implementation of shell script tasks using Linux.

PART-A - ORIGIN AND PHILOSOPHY

❖A1. The Problem This Software Was Created to Solve

Before the emergence of open source software, computer industry was dominated by proprietary software. This type of software never provided the opportunity for users to see and modify the source code. In other words, it theoretically ensured that either users couldn't understand the functionality of program, or they could not modify it. This made the dependency on software companies for any update, bug fix, or addition of features a highly dependent task.
Programming languages like C and C++ were dominating the industry, but they required a high understanding of complex syntax and memory allocation. This had a negative impact on overall productivity, as it required an easy language, which was easy to learn and powerful.
Python is an open-source language, and was created to solve all these issues, while focusing on simplicity and readability. It offered programmers a very concise and readable language to avoid complexities in the code.

On the other hand, Linux offered a free and open-source operating system, which gave complete control over the environment. Linux and Python provided solutions to all the issues faced in system-level and programming-level computing, which were related to the terms "openness, flexibility, and efficiency". Meanwhile Linux offered a free operating system, and users could run any software they wanted.

❖A2. The License — What It Actually Says

Licensing is an important feature of open-source software, as it describes how the software may be utilized, altered, and distributed. The Linux operating system is covered under the GNU General Public License, which guarantees that the software will remain freely and openly accessible to all users. This license permits users to

access the source code, change it, and distribute it, provided that the altered version is also shared with others using the same license.

Python, on the other hand, is covered under the Python Software Foundation License, which is a liberal open-source license. This license allows users to freely utilize, change, and distribute Python, even for business purposes, without imposing limitations. The idea of copyleft, which is a fundamental feature of the GPL, guarantees that the software will remain freely and openly accessible to all users and will not be converted into proprietary software. These licenses are critical as they ensure open and fair nature of software development. They promote collaboration and guarantee that the benefits of the software are shared among all users.

❖A3. The Ethics of Open Source

The open source movement is not only based on technology, but it is also based on good ethical values. The open source software encourages the free availability of knowledge and the power of collaboration to deliver better results.

The ethical values that are followed by the open source includes freedom, transparency, collaboration, and accessibility. Freedom ensures that user is in full control of the software they are using whereas transparency allows the user to look at the source code, which makes the user more comfortable and secure. Collaboration allows developers from all over the world to work together, resulting in fast development of the software and accessibility ensures that the software is available to all, irrespective of their economic status.
The Free Software Foundation and the Open Source Initiative are playing a very important role in the promotion of ethical values. Python is a perfect example of the above-discussed ethical values, as it is developed and maintained by the community all over the world.

PART B — LINUX FOOTPRINT

Linux has emerged as one of the most powerful and widely used operating systems in the modern technological world. Since its creation, it has transformed from a simple open-source operating system to a global thing with a wide variety of uses. The **Linux kernel**, created by Linus Torvalds, is the core of this and provides a stable and secure platform for a variety of operating systems. Its open-source nature allows users to modify and customize it according to their needs, which has been a huge factor in its popularity.
One of the most important domains where Linux has made a huge impact is in field of web servers and internet. Many websites around the world are managed on Linux-based servers due to their reliability, performance, and security. Linux systems are capable of handling large volumes of traffic, making them suitable for enterprise-level applications. Thus, it plays an important role in functioning of the internet, providing services such as websites, applications, and databases.

In addition to web servers, it also plays an important role in cloud computing infrastructure. All main cloud service providers, including Amazon Web Services (AWS), Microsoft Azure, and Google Cloud, heavily rely on Linux-based systems to offer flexible and efficient services to their customers. It provides flexibility to organizations to offer customized solutions and manage their resources efficiently. Linux is also compatible with different programming languages, including Python, which increases its usability in cloud-based applications and services. It is also used in mobile technology infrastructure. The Android mobile operating system used in most mobile devices is based on Linux's kernel code. This shows the flexibility and usability of Linux in different environments, including powerful servers and handheld devices like mobile phones.
Linux is also used in embedded devices and Internet of Things (IoT) devices. Devices like smart TVs, routers, and other devices often use Linux as their operating system. It is also used in other devices like smart home devices and industrial devices because it is light weight and flexible enough to be used in hardware configurations.

Popular Linux platforms like Ubuntu, Debian, and Fedora offer user-friendly interfaces and tools that are specific to the type of user. Ubuntu is known for its simplicity and is often preferred by new users, while Debian is known for its stability, and Fedora is known for offering advanced features.

Another important part of the footprint of Linux is role played by Python programming language in the Linux environment. Python plays a very important role in the Linux environment. It is widely used for automation and development in the Linux system. Many administrative tools are written in Python because it is simpler and more efficient which reduces human involvement and increases productivity. Moreover, footprint of Linux is also seen in the way it reduces the costs associated with the usage of software. Since it is an open-source platform, it does not require the user to pay any licensing fees, which is a major advantage of using Linux operating system. It is also known for offering the highest level of security and is a trusted platform.

In conclusion, the footprint of Linux is wide and varied, covering different areas such as server platforms, cloud computing, mobile gadgets, and embedded devices. The versatility, reliability, and open-source nature of Linux have made it an integral part of technology today. The combination of Python has also boosted the capabilities of Linux.

The following screenshots confirm the installation of Python in the Linux system:

```
shrestha_agarwal@DESKTOP-D2T5Q71:/mnt/c/WINDOWS/system32$ sudo apt install python3
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.12.3-0ubuntu2.1).
python3 set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 63 not upgraded.
shrestha_agarwal@DESKTOP-D2T5Q71:/mnt/c/WINDOWS/system32$
```

PART C — THE FOSS ECOSYSTEM

The Free and Open Source Software (FOSS) ecosystem is a complex network of developers, organizations, tools, and platforms that contribute to development and improvement of open-source software. Contrary to the traditional model of software development, where a single entity dictates the process, the FOSS ecosystem is a decentralized platform. This allows developers from various parts of world to collaborate, share ideas, and contribute to the improvement of open-source software.

One of the main parts of the FOSS ecosystem is the version control systems, specifically Git. Version control systems help developers to track changes in their code, manage various versions of a particular project, and collaborate with other developers. Platforms such as GitHub and GitLab have become essential tools for open-source development. These platforms allow developers to collaborate with other developers, contribute to open-source projects, and develop projects in teams. Another important feature of the FOSS ecosystem is provision of programming languages and tools that aid in development. One programming language that is significant in this case is Python. This is because Python is easy to learn and implement in application development, among other factors. It is used in web development, data science, machine learning and many other fields. It is also efficient in application development due to its large number of libraries, including NumPy, Pandas, and TensorFlow.

Python also attracts an important global community that actively participates in its development. This is because members of this community actively contribute to its development and evolution. This is through contributing libraries, correcting bugs, improving documentation, and offering tutorials. This ensures Python's continued evolution and development to accommodate current technological changes. The provision of learning materials also ensures that new participants learn Python and actively participate in its development as an open-source system.

The FOSS ecosystem encourages innovation since there are no barriers to entry. This is due to the fact that most tools and software are free. As a result, there are startups and innovations using open-source platforms.

Another aspect of FOSS ecosystem is the great influence it has on education and learning. Learners and programmers have access to open-source tools for learning. Python is a good example of a programming language used for learning. It is easy to learn and has many applications and is therefore, suitable for beginners. In conclusion, the FOSS ecosystem is a great platform for innovation, learning, and development.

PART D - OPEN SOURCE VS PROPRIETARY

DIMENSION	PYTHON (OPEN SOURCE)	PROPRIETARY SOFTWARE
Cost	It is free to use, no licensing fees is required.	Licensing fees or subscription is required.
Security	Source code is open and can be audited by anyone	Source code is closed, users must trust the company.
Support	Community support, forums, documentation	Official paid support from company.
Freedom to Modify	Complete freedom to modify and customize	No freedom to modify source code.
Community Vs Corporate Control	Community-driven development.	Controlled by a single company.
Flexibility	Highly flexible and adaptable.	Limited flexibility.

Python is absolutely free, and is open-source, which means that it does not require any subscription fees, license fees, or other charges. Anyone can download, use, and distribute Python without paying any charges. This is not the case with other programming software, as they often require paid licenses or subscription fees. These fees can add up over time, especially for students or small organizations.

Another significant advantage of using Python is its openness. Since the source code is made publicly available, developers can view it, understand it, and modify it if they want to. This is particularly advantageous from the security perspective, where vulnerabilities can be addressed by the entire world. In other words, it is not just the developers who are responsible for security but the entire world. Moreover, other programming software is closed-source, which means that the user cannot view the source code so they have to depend totally on the company.

In addition, Python has an enormous advantage over other programming software with regard to support. Python developers can access support from GitHub, Stack Overflow, or other forums. This is an extremely effective support system. In addition, it is free. On the other hand, other programming software offers support services, but they are paid, and the user can only access them according to the company's policies.

SHELL SCRIPTS

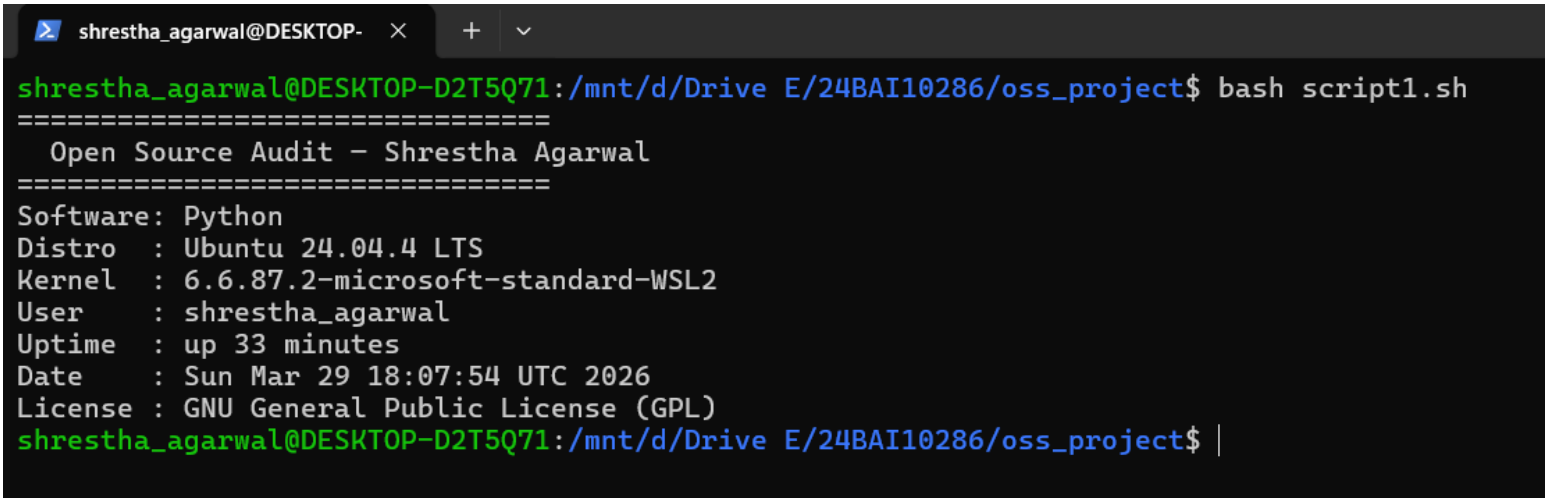
Script 1 — System Identity Report

Purpose:

The purpose of this script is to create a detailed system identification report while offering a quick look at the system environment, which is helpful for auditing and documentation.

Concepts Used:

This script uses basic shell scripting features such as variables, command substitution and system commands like *uname*, *whoami* , *uptime*, and *date* to get system information. It reads */etc/os-release* for OS details and uses *echo* to produce formatted output.



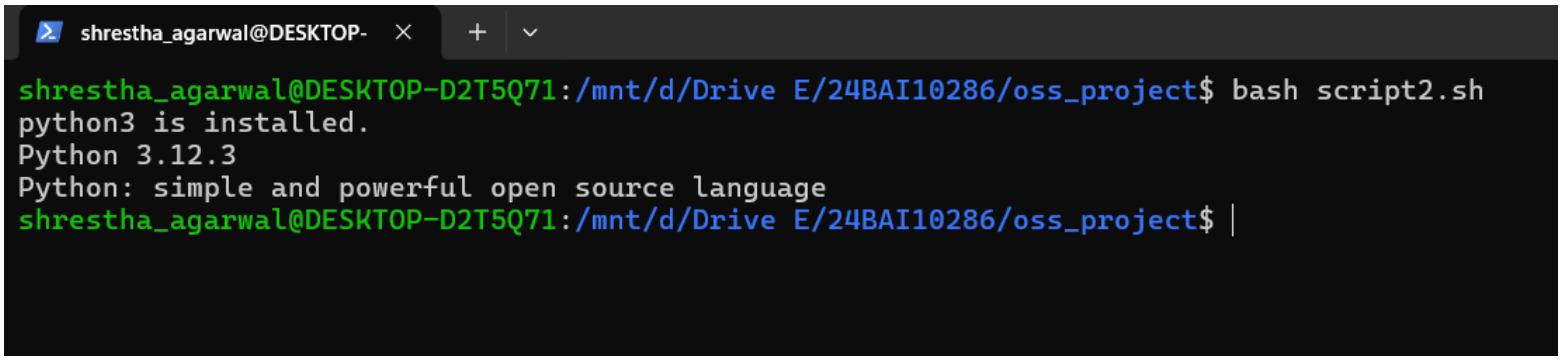
Script 2 — Python Package Inspector

Purpose:

This script checks if Python is installed on the system. It also shows its version and basic details. This ensures that the necessary tools are available before starting development.

Concepts Used:

This script uses conditional statements like if-else and case, along with command -v, to check if a package is installed. It shows how to make decisions and branch in shell scripting.



Script 3 — Disk and Permission Auditor

Purpose:

The purpose of the script is to analyze system directories by checking how much disk space they use and what permissions they have. It also checks presence of Python binary and helps with monitoring and auditing the system.

Concepts Used:

This script checks directories, their sizes and permissions using arrays, loops, and file handling. It also uses conditions to check for directory existence, which shows how well shell scripts can automate system checks.

```
shrestha_agarwal@DESKTOP-
shrestha_agarwal@DESKTOP-D2T5Q71:/mnt/d/Drive E/24BAI10286/oss_project$ bash script3.sh
Directory Audit Report
----------------------
/etc => Permissions: drwxr-xr-x root root | Size: 4.3M
/var/log => Permissions: drwxrwxr-x root syslog | Size: 17M
/home => Permissions: drwxr-xr-x root root | Size: 40K
/usr/bin => Permissions: drwxr-xr-x root root | Size: 95M
/tmp => Permissions: drwxrwxrwt root root | Size: 24K

Python Config Directory:
drwxr-xr-x 2 root root 4096 Feb 10 00:55 /etc/python3
shrestha_agarwal@DESKTOP-D2T5Q71:/mnt/d/Drive E/24BAI10286/oss_project$
```

Script 4 — Log File Analyzer

Purpose:

The purpose of the script is to analyze the log files to find how many times a keyword, like "error," appears and shows the most recent matches. It makes it easier to find the issues in our file quickly for debugging and monitoring.

Concepts Used:

This script uses the concept of loops, file handling, and grep to find patterns and to analyze log files. It counts how many times something happens, takes inputs, and shows how shell scripting can help with text processing.

```
shrestha_agarwal@DESKTOP-
shrestha_agarwal@DESKTOP-D2T5Q71:/mnt/d/Drive E/24BAI10286/oss_project$ bash script4.sh /var/log/dpkg.log error
Keyword 'error' found 39 times in /var/log/dpkg.log

Last 5 matching lines:
2026-02-10 00:56:02 status unpacked liberror-perl:all 0.17029-2
2026-02-10 00:56:32 configure liberror-perl:all 0.17029-2 <none>
2026-02-10 00:56:32 status unpacked liberror-perl:all 0.17029-2
2026-02-10 00:56:32 status half-configured liberror-perl:all 0.17029-2
2026-02-10 00:56:32 status installed liberror-perl:all 0.17029-2
shrestha_agarwal@DESKTOP-D2T5Q71:/mnt/d/Drive E/24BAI10286/oss_project$
```

<u>OBSERVATION :</u> Sometimes matches may not represent actual system errors as keyword may appear inside packages names. This limitation of keyword based analysis suggests the requirement of more advance filtering techniques

Script 5 — Open Source Manifesto Generator

Purpose:

The purpose of this script is to make a unique open-source manifesto based on what the user types in. It gathers answers and makes a structured message to share thoughts on the philosophy of open source.

Concepts Used:

This script makes dynamic content by using user input, variables, and file handling. It writes answers to a file while it also includes the current date, showing how users interact with shell scripts.

```
shrestha_agarwal@DESKTOP-

shrestha_agarwal@DESKTOP-D2T5Q71:/mnt/d/Drive E/24BAI10286/oss_project$ bash script5.sh
Answer three questions to generate your manifesto.

1. Name one open-source tool you use every day: Python
2. In one word, what does 'freedom' mean to you? travel
3. Name one thing you would build and share freely: songs

Manifesto saved to manifesto_shrestha_agarwal.txt

Open Source Manifesto - 29 March 2026

I use Python every day. To me, freedom means travel.
I believe in sharing knowledge openly. I would like to build songs and share it freely with the world.
shrestha_agarwal@DESKTOP-D2T5Q71:/mnt/d/Drive E/24BAI10286/oss_project$
```

PROJECT TIMELINE

The project was completed in a structured manner.

- First, the basic concepts of Linux and FOSS were studied to create a strong base.
- After this, the concepts of Python and shell scripting techniques were learned.
- In the next step, the scripts were written and implemented to check for proper functionality.
- Finally, the report was prepared by structuring the theoretical and practical knowledge in a proper manner.

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- GeeksforGeeks. Linux and Python Tutorials. https://www.geeksforgeeks.org

CONCLUSION

In conclusion, it can be seen that the project gave a complete understanding of Linux and Free and Open Source Software (FOSS) and their importance in the current era of computing. It explained that open source projects support transparency, collaboration and innovation through which users can access and modify source code. Through study of Linux and Python, we saw how open source technologies have transformed software development and system operations. With the help of analysis of licensing, ethical issues and real world applications we saw that FOSS plays an important role in cost reduction and access improvement.

The practical implementation through use of shell scripts helped us in understanding the study of automation and system management in the Linux environment. Also, in this project we not only learned about theoretical concepts but also saw growth in practical applications of open source tools like Python and Linux.